



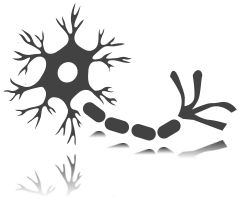
Dynamical neural change in visual cortex during learning

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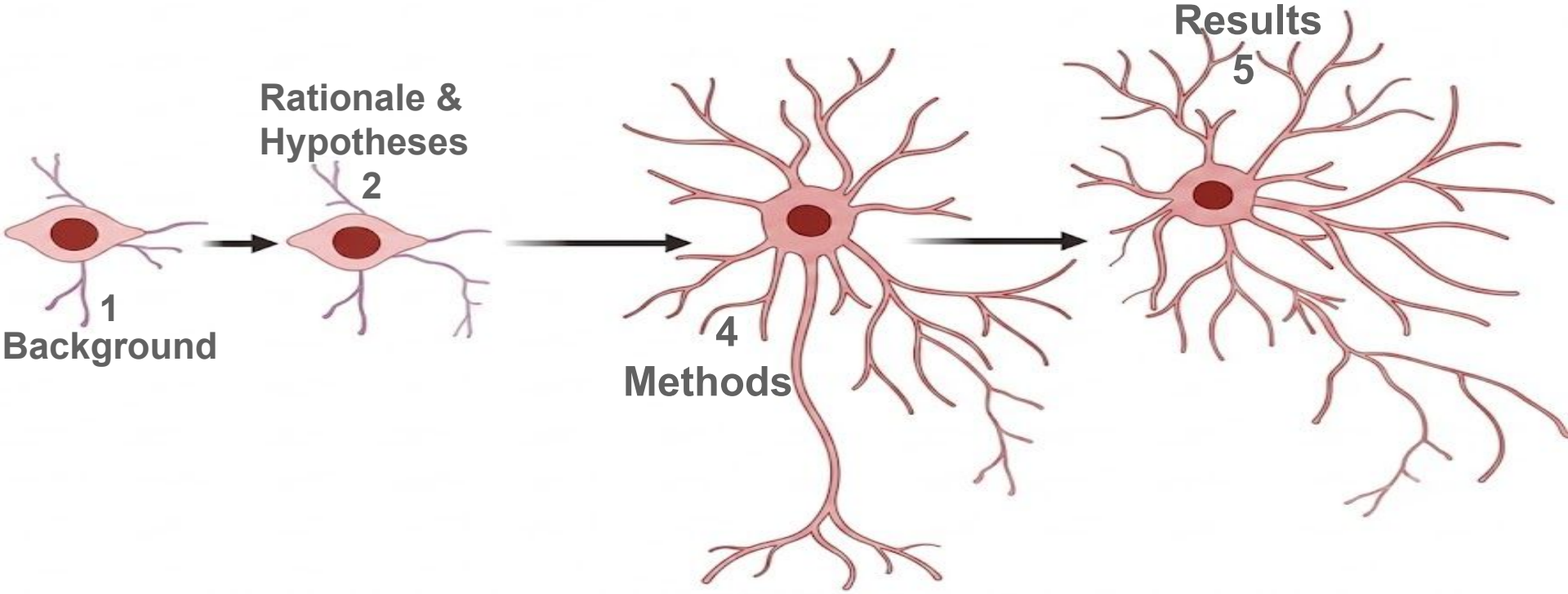
TAs: Lina Alqam, Amalya Hakobyan

Pod: Tokoloshe Black Cumin Group: Ratatouille





Outlines



Representation Learning in the Visual Cortex

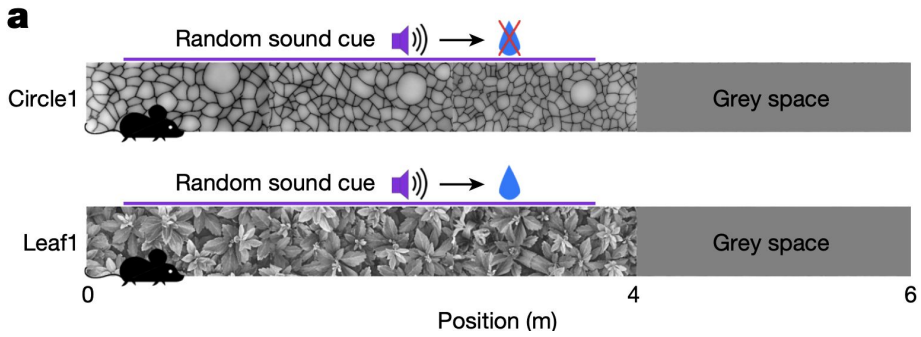


The Known Baseline

- The Established Phenomenon (Jurjut et al., 2017): Reward-associated learning enhances neural representations in the visual cortex.
- Increases neural discriminability and stimulus selectivity, often before measurable behavioral improvements appear.

The Experiment (Zhong et al., 2025): Investigated how visual experience and rewards actively shape these neural representations.

- Mice navigated virtual "leaf" and "circle" corridors under two conditions:
- **Supervised Group**: Received a sugar water reward in the leaf corridor.
- **Unsupervised Group**: Explored the exact same environments without any reward.



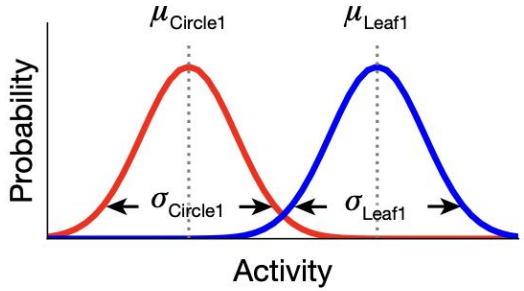
Neural selectivity can increase without reward

Neural Discriminability (d')

- measures how well the neuron(s) distinguishes between two different stimuli (leaf and circle).

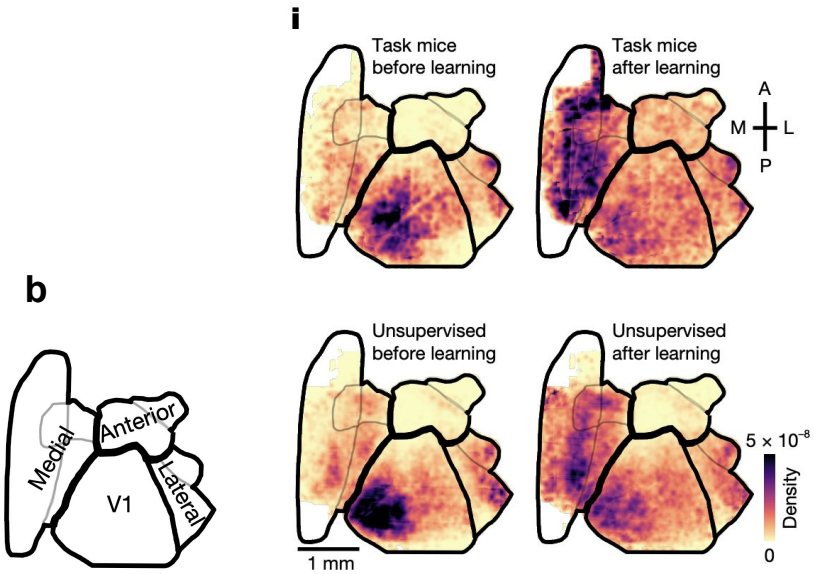
a

$$d' = \frac{\mu_{\text{Leaf1}} - \mu_{\text{Circle1}}}{(\sigma_{\text{Leaf1}} + \sigma_{\text{Circle1}})/2}$$



Key finding

- Most representational changes occurred even **without reward**.



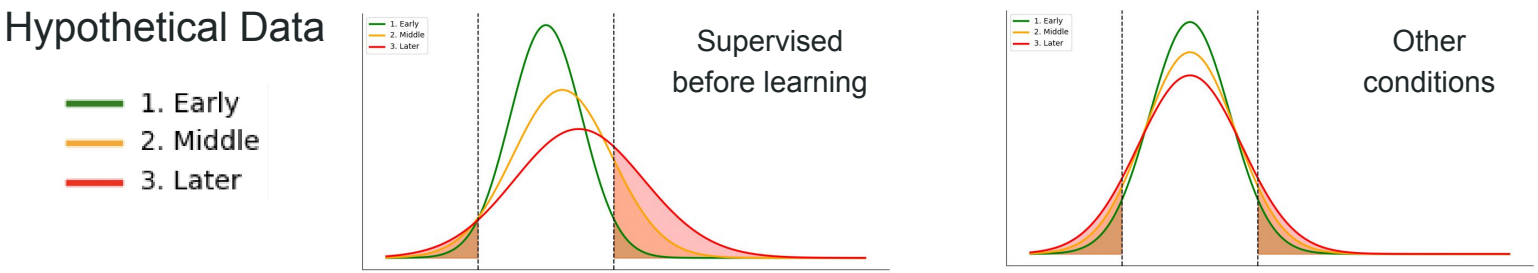
Rationale & Hypotheses

Rationale

- How neural representations evolve trial-by-trial within a single session remains a mystery.
- We investigate intra-session changes in the population d′ distribution.

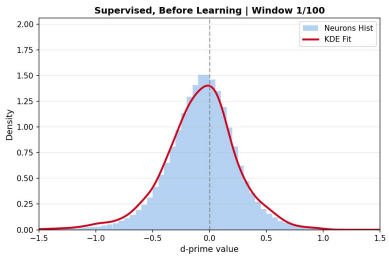
Hypothesis: different dynamics in different conditions

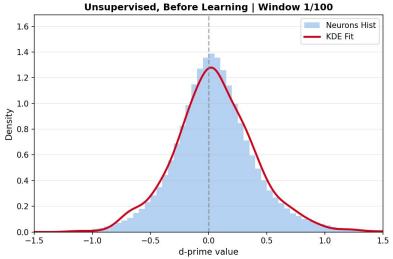
1. Central Tendency (Median, H1)
2. Population Dispersion (Std Dev σ, H2)
3. Fraction of selective neurons (Fraction |d'| ≥ 0.3, H3)

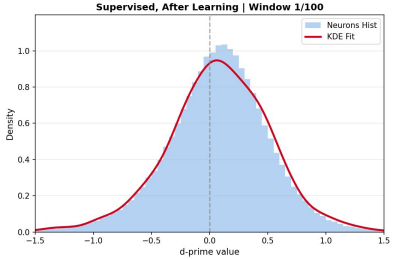


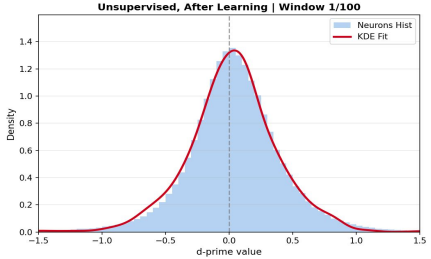
Methods

- Reconstructed neuronal responses of mice (*N* = 13) from visual cortical **SVD components**, and evaluated **d' distributions** across trials using **sliding window.**







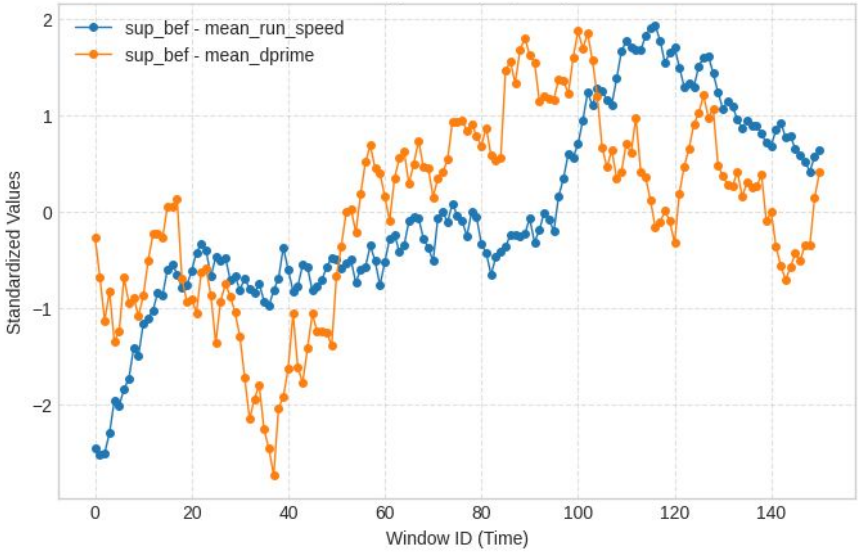


- Implemented **Generalized Additive Mixed Models (GAMMs)** to capture non-linear trajectories of d' distributions across session time

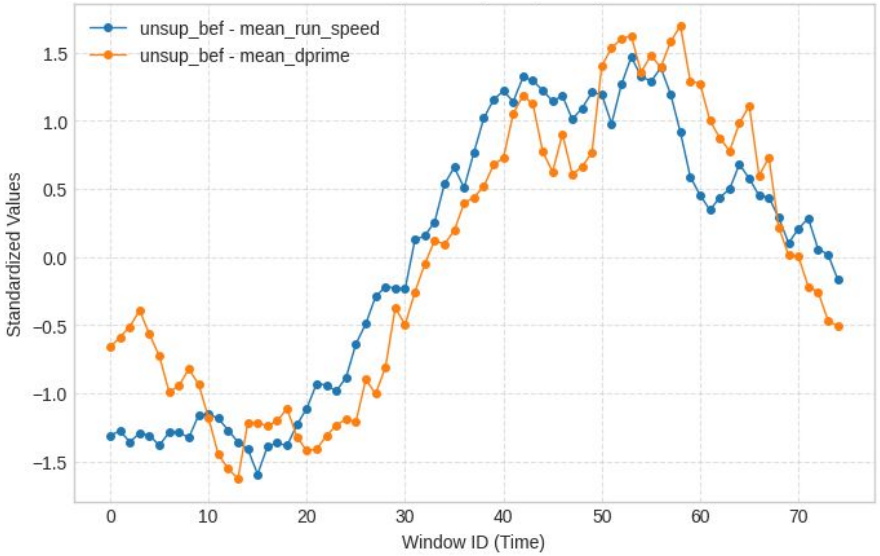
$$Y_{m,i} = \beta_0 + \sum_{k=1}^{K} \beta_k \text{Condition}_{k,m,i} + \sum_{k=1}^{K} f_k(t_{m,i}) \text{Condition}_{k,m,i} + \gamma \text{Speed}_{m,i} + b_m + \epsilon_{m,i}$$

Running Speed Affects Performance

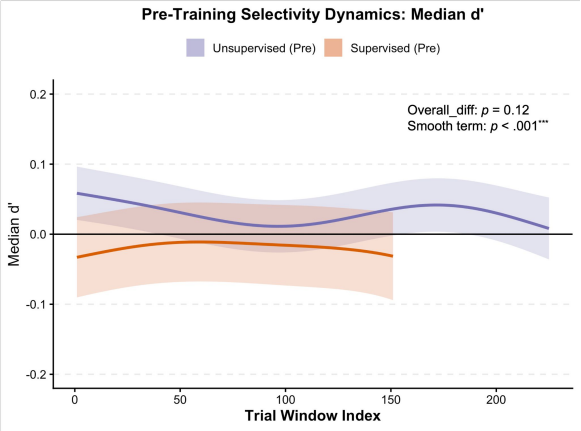
Supervised mouse before learning

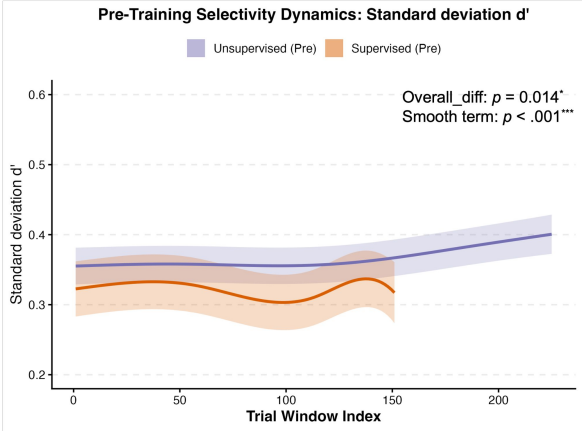


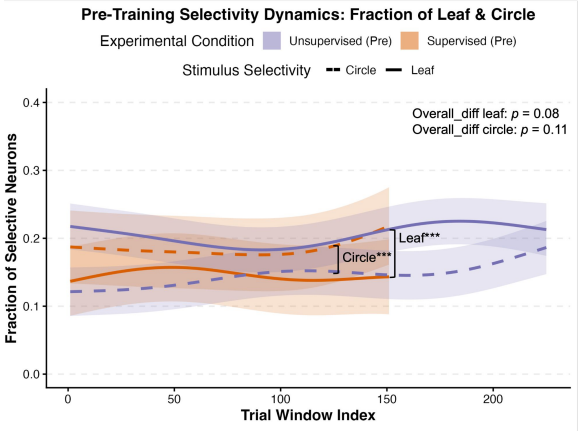
Unsupervised mouse before learning



Pre-training: Similar beginnings, divergent paths

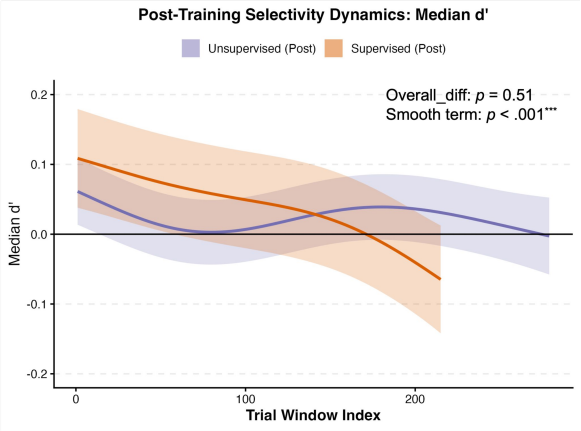


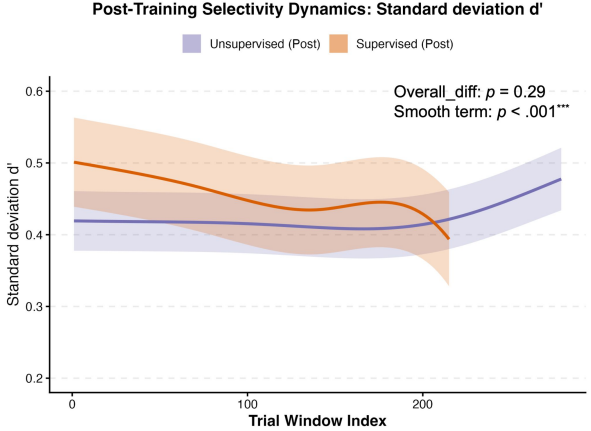


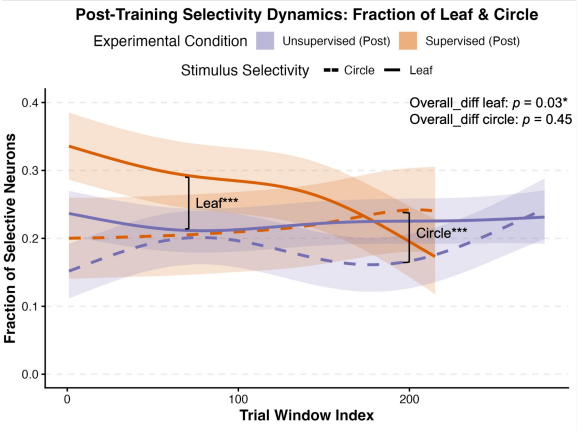


- Median d′: supervised (orange) and unsupervised (purple) overlap, no clear separation
- Std dev: significant group difference (p = .014), unsupervised trending higher
- Selectivity: unsupervised pulls ahead on leaf & circle selectivity by window 150

Post-training: Reward yields preference



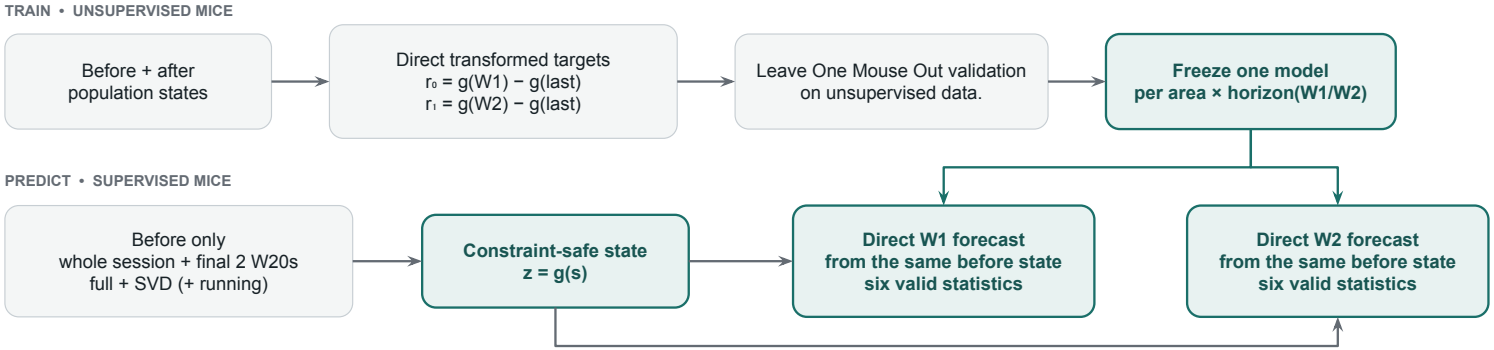




- Only **Leaf-selective neurons** differed significantly before groups in post-training session, suggesting **target-specific plasticity**.
- Pre–post comparisons showed stable median d′ and circle selectivity trajectory, alongside ongoing within-session dynamics.

Bonus: d’ predictions from unsupervised to supervised

Using unsupervised (before/after) data, predict region-wise d-prime stats for W1/W2 after learning for supervised mice by using input from supervised (before) data.



State and constraint-safe transformation

$$s = (q_{05}, q_{50}, q_{95}, \mathrm{SD}, \mathrm{pL}, \mathrm{pC}), \quad p_0 = 1 - \mathrm{pL} - \mathrm{pC}$$

$$z = g(s) = \left(q_{50},\ \ln(q_{50} - q_{05}),\ \ln(q_{95} - q_{50}),\ \ln \mathrm{SD},\ \ln(\mathrm{pL}/p_0),\ \ln(\mathrm{pC}/p_0) \right)$$

g^{-1} uses exponentiations, g is used to add non-linearity and a constraint safe transformation for ridge regression to operate on.

Model Family

Candidate family: persistence (Δz = 0) • mean shift (β = 0) • ridge GLM

Ridge = six independent GLMs per mouse area per horizon (W1/W2)

$$\hat{\mathbf{s}}_{\text{after},\,h} = g^{-1}\left(g(\mathbf{s}_{\text{before}}) + \hat{\boldsymbol{\Delta}}_h \right)$$

after-state = before-state + learned boundary shift → six valid statistics (h = W1, W2)

$$\hat{\boldsymbol{\Delta}}_h = \boldsymbol{\beta}_{0,h} + \mathbf{B}_h \mathbf{x}_{\text{before}}$$

ridge GLM (6 outputs) **selected: $\mathbf{B}_h = 0$ ⇒ $\hat{\boldsymbol{\Delta}}_h$ = a constant exposure shift**

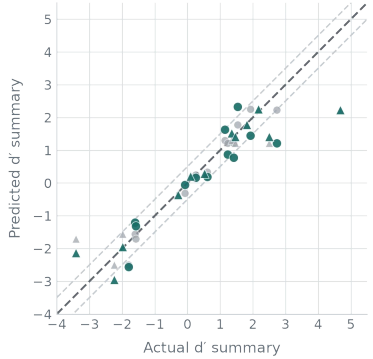
Area specific d’ predictions: Results

Frozen supervised transfer • each point is one actual versus predicted area-summary statistic

Dashed center = exact match; light dashed lines = ±0.5 d′

Direct model Persistence W1 W2

V1*

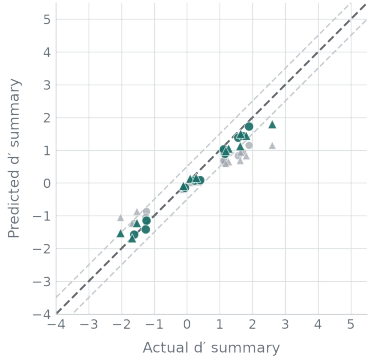


Within ±0.5 d’
Model 66.7% · Persistence 83.3%

MAE: Model 0.523 d’ · Persistence 0.406 d’

pLeaf/pCircle MAE: Model 6.2 vs persistence 6.0 %

Medial (mHV)

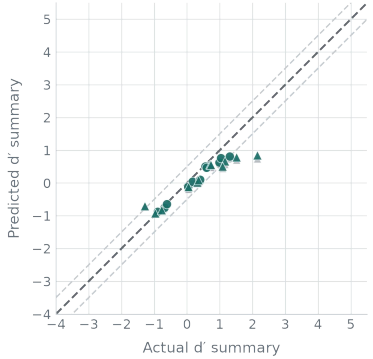


Within ±0.5 d’
Model 91.7% · Persistence 50.0%

MAE: Model 0.213 d’ · Persistence 0.542 d’

pLeaf/pCircle MAE: Model 5.6 vs persistence 9.3 %

Anterior (aHV)

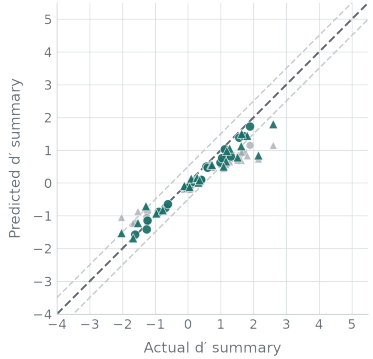


Within ±0.5 d’
Model 83.3% · Persistence 75.0%

MAE: Model 0.285 d’ · Persistence 0.337 d’

pLeaf/pCircle MAE: Model 12.0 vs persistence 15.8 %

Overall (mHV+aHV)



Within ±0.5 d’
Model 87.5% · Persistence 62.5%

MAE: Model 0.249 d’ · Persistence 0.439 d’

pLeaf/pCircle MAE: Model 8.8 vs persistence 12.6 %

*** V1 is SVD-only, mHV/aHV used full neural data, d’ is a 4-tuple (q05, median, q95, SD), persistence baseline = just use current window’s stats**

References

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Thank you for listening!

